EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A public key must be kept secret to use public-key encryption.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

DO NOW — RECALL

Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

1. Which technique protects data against casual interception?

2. What does a network proxy do?

3. What is the difference between confidentiality and integrity?

4. Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Encryption.

2. Model answer

Acts as an intermediary for requests/responses.

3. Model answer

Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

4. Model answer

Q changes to 1 on the next rising edge, not simply when D changes.

Vocabulary — know these terms

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

TLS — Protocol establishing secure authenticated channels.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe symmetric and asymmetric key arrangements. [6]

Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

NEW LEARNING

Read, annotate and explain

Understand

Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Trace

Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Evaluate

Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Predict first

Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

Show method

Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve